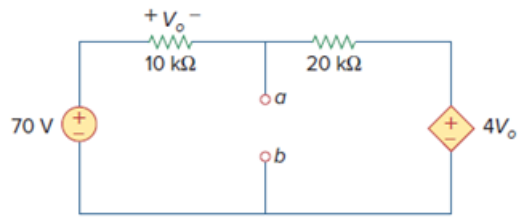


Problem

Find the Thevenin equivalent at terminals a - b of the circuit in Fig. 4.107.

Figure. 4.107:

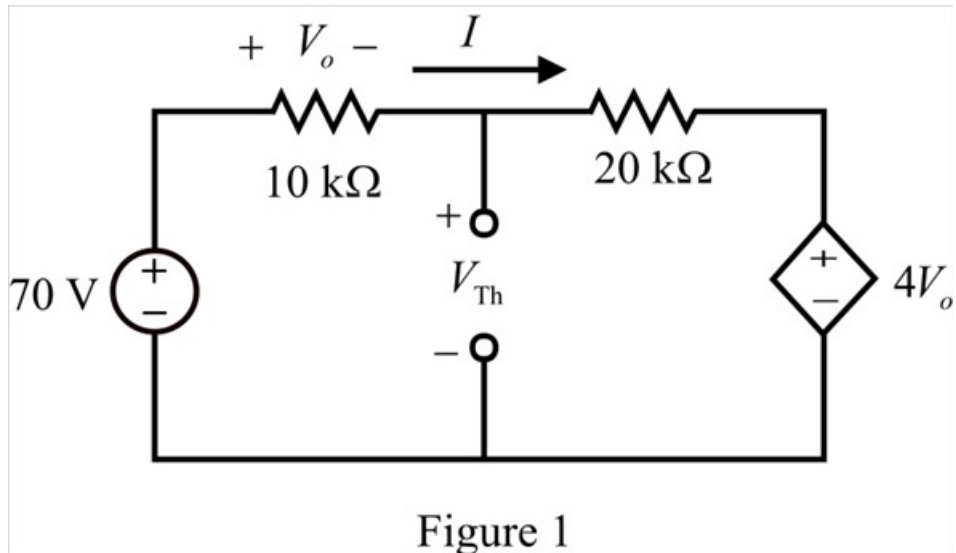


Step-by-step solution

Step 1 of 7

Refer to the circuit diagram depicted in Figure 4.107 in the textbook.

Draw the circuit diagram notating with loop current and Thevenin voltage.



Step 2 of 7

From the Figure 1, the value of voltage V_o is,

$$V_o = 10 \times 10^3 I$$

Apply Kirchhoff's voltage law around the loop.

$$-70 + 10 \times 10^3 I + 20 \times 10^3 I + 4V_o = 0$$

$$30 \times 10^3 I + 4V_o = 70$$

$$30 \times 10^3 I + 4(10 \times 10^3 I) = 70 \quad (\text{since } V_o = 10 \times 10^3 I)$$

$$70 \times 10^3 I = 70$$

$$I = 1 \text{ mA}$$

Apply Kirchhoff's voltage law to the left side of the circuit.

$$-70 + 10 \times 10^3 I + V_{Th} = 0$$

$$V_{Th} = 70 - 10 \times 10^3 (1 \times 10^{-3})$$

$$= 70 - 10$$

$$= 60 \text{ V}$$

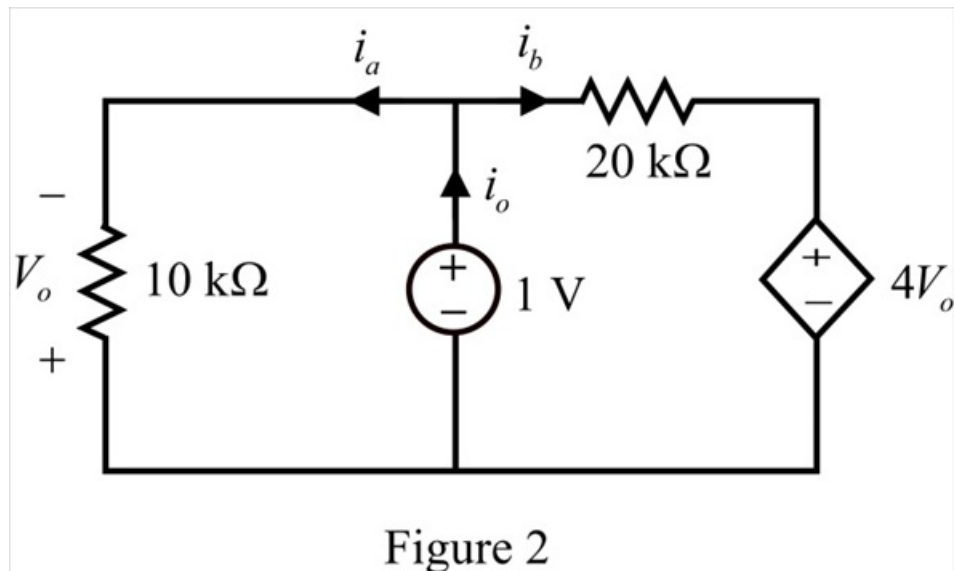
Therefore, the value of Thevenin voltage, V_{Th} is 60 V.

Step 3 of 7

Since the circuit has dependent source, to find the Thevenin equivalent resistance, place a 1 V test source across the open terminals a and b . Then, determine the current supplied by the test source, thereafter; find the Thevenin equivalent resistance by using the following formula.

$$R_{Th} = \frac{1 \text{ V}}{i_o}$$

Draw the modified circuit diagram.



[Comments \(1\)](#)



Anonymous

why did the sign get flipped on v0

Step 4 of 7

From the Figure 2, the value of voltage V_o is,

$$V_o = -1 \text{ V}$$

Calculate the value of current i_a .

$$\begin{aligned} i_a &= \frac{-V_o}{10 \times 10^3} \\ &= \frac{1}{10 \times 10^3} \\ &= 0.1 \text{ mA} \end{aligned}$$

Apply Kirchhoff's voltage law to the outer loop of the circuit in Figure 2.

$$\begin{aligned} -10 \times 10^3 i_a + 20 \times 10^3 i_b + 4V_o &= 0 \\ -10 \times 10^3 (0.1 \times 10^{-3}) + 20 \times 10^3 i_b + 4(-1) &= 0 \\ -1 + 20 \times 10^3 i_b - 4 &= 0 \\ 20 \times 10^3 i_b &= 5 \\ i_b &= 0.25 \text{ mA} \end{aligned}$$

Step 5 of 7

Calculate the total current delivered by the 1 V test source.

$$\begin{aligned} i_o &= i_a + i_b \\ &= 0.1 \text{ mA} + 0.25 \text{ mA} \\ &= 0.35 \text{ mA} \end{aligned}$$

Step 6 of 7

Determine the Thevenin equivalent resistance.

$$\begin{aligned} R_{Th} &= \frac{1 \text{ V}}{i_o} \\ &= \frac{1 \text{ V}}{0.35 \times 10^{-3} \text{ A}} \\ &= 2857.143 \text{ } \Omega \\ &= 2.86 \text{ k}\Omega \end{aligned}$$

Therefore, the value of Thevenin resistance, R_{Th} is 2.86 kΩ.

Step 7 of 7

Draw the Thevenin equivalent circuit across the terminals a - b .

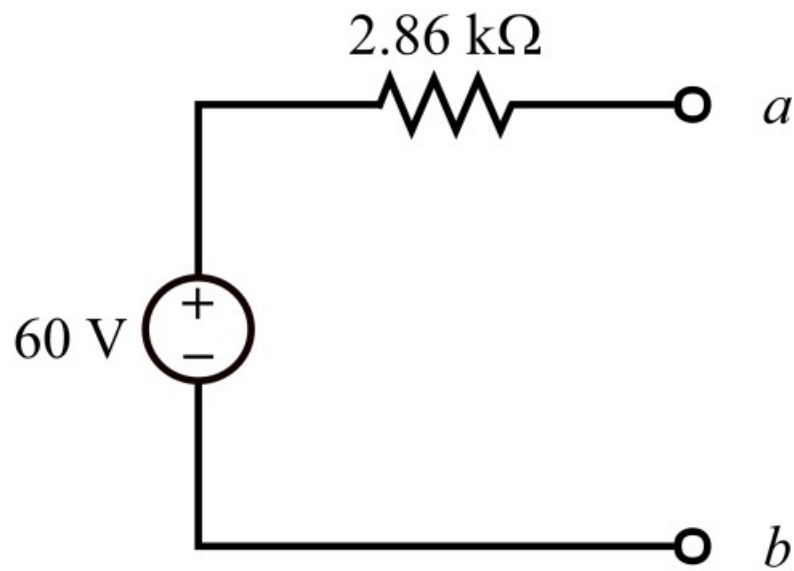


Figure 3

Hence, the Thevenin equivalent circuit is drawn.